

Welfare Theorems

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Heuristically

- First Welfare Theorem: Every Walrasian Equilibrium (WE) is Pareto Optimal (PO).
- Second Welfare Theorem: If preferences and technology are convex, every PO allocation can be supported as a WE after an appropriate redistribution of endowments.

First Welfare Theorem

First Welfare Theorem

Theorem 1 (First Welfare Theorem) Let (x^*, y^*, p^*) be a WE of the Arrow-Debreu economy $\mathcal{E}$. If preferences are, complete, transitive, and locally nonsatiated then (x^*, y^*) is PO.

Proof. We begin with two implications of preference maximization which we label (a) and (b).

Implication (a)

Claim. It follows from preference maximization that

$$x'_i \succ_i x_i^* \implies p^* \cdot x'_i > p^* \cdot x_i^*$$

To prove this, arguing by contradiction:

- Suppose $x'_i \succ_i x_i^*$ and $p^* \cdot x'_i \leq p^* \cdot x_i^*$.
- Since $x_i^* \in B_i \wedge p^* \cdot x'_i \leq p^* \cdot x_i^*$ then $x'_i \in B_i(p^*, y^*)$
- Since $x_i^* \in D_i(p^*, y^*)$, $(\forall x_i \in B_i(p^*, y^*)) x_i^* \succeq_i x_i$; a contradiction.

Implication (b)

Claim. $x'_i \succeq_i x_i^* \implies p^* \cdot x'_i \geq p^* \cdot x_i^*$

- By contradiction: Suppose $x'_i \succeq_i x_i^* \wedge p^* \cdot x'_i < p^* \cdot x_i^*$.
- $\exists \epsilon > 0 : (\forall z, \|z - x'_i\| < \epsilon) p^* \cdot z < p^* \cdot x_i^*$ (by continuity of "dot product".)
- LNS $\implies \exists x''_i \in X_i : \|x''_i - x'_i\| < \epsilon, x''_i \succ_i x'_i$.
- Then, $p^* \cdot x''_i < p^* \cdot x_i^* \wedge x''_i \succ_i x'_i \succeq_i x_i^*$. Thus $x''_i \succ_i x_i^*$ (by completeness, transitivity $\succeq_i$; needs proof).
- Then $x_i^* \notin D_i(p^*, y^*)$, a contradiction.



Lemma. Suppose preferences $\succeq_i$ are transitive and complete. Then

$$x_i'' \succ_i x_i' \succeq_i x_i^* \implies x_i'' \succ x_i^*.$$

Proof.

- By transitivity $x_i'' \succeq_i x_i^*$.
- Suppose $x_i'' \not\succeq_i x_i^*$. By completeness, $x_i^* \succeq_i x_i''$.
- Contradiction: $x_i' \succeq_i x_i^*$ and $x_i^* \succeq_i x_i''$. Then by transitivity $x_i' \succeq_i x_i''$.
- We have shown (a) and (b): If a bundle is at least as desirable it cannot be cheaper; if it is more desirable it is more expensive than the demand bundle.

Proof of First Welfare theorem

Arguing by contradiction, suppose exists an allocation (x', y') that Pareto dominates (x^*, y^*) , i.e.

$$\begin{aligned} \exists(x', y') \mid & \sum_{i=1}^I x'_i = \bar{\omega} + \sum_{j=1}^J y'_j, \\ & \forall i, x'_i \succeq_i x_i^* \text{ and} \\ & \exists i, x'_i \succ_i x_i^* \end{aligned}$$

$$(a) \wedge (b) \implies \sum_{i=1}^I p^* \cdot x'_i > \sum_{i=1}^I p^* \cdot x_i^*$$

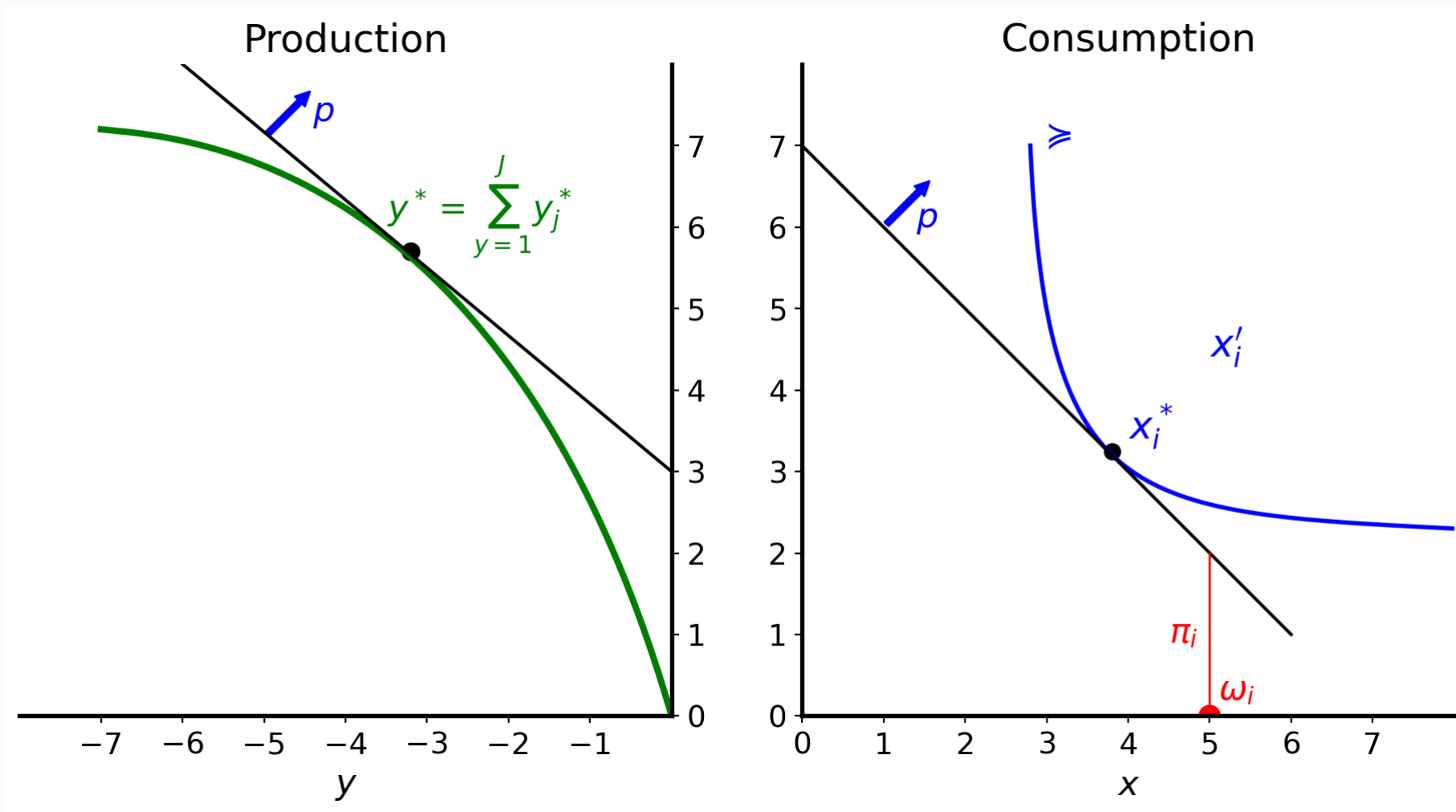
$$\begin{aligned}
\sum_{i=1}^I \sum_{\ell=1}^L p_{\ell}^* x'_{i\ell} &> \sum_{i=1}^I \sum_{\ell=1}^L p_{\ell}^* x^*_{i\ell} \\
\sum_{\ell=1}^L \sum_{i=1}^I p_{\ell}^* x'_{i\ell} &> \sum_{\ell=1}^L \sum_{i=1}^I p_{\ell}^* x^*_{i\ell} \\
\sum_{\ell=1}^L p_{\ell}^* \sum_{i=1}^I x'_{i\ell} &> \sum_{\ell=1}^L p_{\ell}^* \sum_{i=1}^I x^*_{i\ell}
\end{aligned}$$

- (i.e., in aggregate, x'_i must be more expensive than x_i^*)

$$\begin{aligned}
p^* \cdot \sum_{i=1}^I x'_i &> p^* \cdot \sum_{i=1}^I x_i^* \\
&= p^* \cdot \bar{\omega} + p^* \cdot \sum_{j=1}^J y_j^* \\
&\geq p^* \cdot \bar{\omega} + \sum_{j=1}^J (p^* \cdot y'_j) = p^* \cdot \sum_{i=1}^I x'_i
\end{aligned}$$

(by profit maximization)

First Welfare Theorem, remarks



Without LNS

To interpret FWT as endorsement of market economies, additional assumptions are necessary.

- Complete markets: there are markets for all goods.
- WE is the outcome of a market economy (this includes the price-taking assumption).
- No externalities.
- FWT says nothing about distribution. A PO allocation may imply very little utility for many consumers.

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- May fail in overlapping generations economies Samuelson (1958). Good summary in Weil (2008).

When there are double infinity of agents and goods, the order of summation cannot be changed.

As a technical reference see, for instance, Fubini's theorem, Rudin (1987), Theorem 8.8, and counterexample 8.9. See also Tonelli's Theorem.

References

Rudin, Walter. 1987. *Real and Complex Analysis*. 3rd ed. McGraw-Hill.

Samuelson, Paul Antony. 1958. "An Exact Consumption-Loan Model of Interest with or Without the Social Contrivance of Money." *Journal of Political Economy* 66 (6): 467–82.

Weil, Philippe. 2008. "Overlapping Generations and the First Welfare Theorem." *Journal of Economic Perspective* 22 (4): 115–34.